

- **Under-stabilization (δ too small)** Insufficient artificial diffusion is injected, allowing oscillations.
- **Over-stabilization (δ too big)** Excessive stabilization introduces an overabundance of artificial diffusion, fundamentally altering the physics of the original governing equation

Our ideal values stand in the middle.

3 project 3: Parabolic equation

3.1 Problem Formulation

We consider the domain

$$\Omega = [-1, 1]^2 = \{(x, y) \in \mathbb{R}^2 : -1 \leq x, y \leq 1\}$$

and the parabolic equation with Dirichlet boundary conditions:

$$\frac{\partial u}{\partial t} = \operatorname{div}(D \nabla u) + f(x, t) \quad \text{for } (x, t) \in \Omega \times [0, T], \quad (19)$$

$$u(x, 0) = u_0(x) \quad \text{for } x \in \Omega, \quad (20)$$

$$u(x, t) = 0 \quad \text{for } t > 0, x \in \partial\Omega. \quad (21)$$

We are given the manufactured solution $u(x, y, t) = e^{-t} \sin(\pi x) \cos(\frac{\pi}{2} y)$ and the terminal time $T = 1$. We Assume $D = 1$

We formulate the weak form of the parabolic problem by seeking $u \in L^2(0, T; H_0^1(\Omega))$ with time derivative $\dot{u} := \partial_t u \in L^2(0, T; H^{-1}(\Omega))$ such that:

$$(\dot{u}, v)_{L^2(t)} + a(u, v)(t) = (f, v)_{L^2(t)} \quad \forall v \in H_0^1(\Omega), \text{ for a.e. } t \in [0, T],$$

We then introduce a conforming $\mathcal{P}^1$ Galerkin approximation using the finite element space $\mathcal{V}_h = \operatorname{span}\{\phi_1, \dots, \phi_{N_h}\}$. We seek an approximate solution of the form:

$$u_h(x, y, t) = \sum_{j=1}^{N_h} C_j(t) \phi_j(x, y)$$

such that testing against any basis function ϕ_i yields:

$$\sum_{j=1}^{N_h} \dot{C}_j(t) (\phi_j, \phi_i)_{L^2} + \sum_{j=1}^{N_h} C_j(t) a(\phi_j, \phi_i) = (f(t), \phi_i)_{L^2} \quad \forall i = 1, \dots, N_h.$$

This can be formulated compactly in matrix form as a system of Ordinary Differential Equations (ODEs):

$$P \dot{u}(t) + A u(t) = b(t) \quad \text{for } t \in [0, T], \quad (22)$$

where $u(t) = (C_j(t))$ is the vector of nodal unknowns, and the mass matrix $P = (p_{ij})$, stiffness matrix $A = (a_{ij})$, and load vector $b(t) = (b_i(t))$ are defined by:

$$p_{ij} = (\phi_j, \phi_i)_{L^2}; \quad a_{ij} = a(\phi_j, \phi_i); \quad b_i(t) = (f(t), \phi_i)_{L^2}.$$

This represents a semi-discrete formulation, in contrast to the fully stationary discrete equations solved in the previous two projects. To obtain a fully discrete numerical approximation of $u_h(T)$, we apply the θ -method for time integration. Partitioning the interval $[0, T]$ uniformly into $\{t_0, t_1, \dots, t_N\}$ where $t_n = n\Delta t$, and setting $u^n \approx u(t_n)$, we rearrange the terms to solve for the unknown state at t_{n+1} .

$$(P + \theta \Delta t A) u^{n+1} = (P - \Delta t(1 - \theta)A) u^n + \theta b^{n+1} + (1 - \theta)b^n.$$

3.2 Implementation of solver

$$p_{ij} = (\phi_j, \phi_i)_{L^2(\Omega)}$$

$$\int_T \phi_i^\alpha \phi_j^\beta \phi_k^\gamma dx = \frac{2\alpha!\beta!\gamma!}{(\alpha + \beta + \gamma + 2)!} |T| \quad (23)$$

where $|T|$ is the area of the triangle. For the local mass matrix entries, the exponents are restricted to $\alpha + \beta \leq 2$ (with $\gamma = 0$).

- $(i = j)$: Setting $\alpha = 2, \beta = 0$, the integral evaluates to $\frac{2(2!)}{4!} |T| = \frac{|T|}{6}$.
- $(i \neq j)$: Setting $\alpha = 1, \beta = 1$, the integral evaluates to $\frac{2(1!)(1!)}{4!} |T| = \frac{|T|}{12}$.

Consequently, the local mass matrix for any element T can be explicitly written as:

The overall execution of the θ -method is divided into two primary stages:

- **Pre-processing:** The spatial matrices A (stiffness) and P (mass) are assembled only once since the domain geometry is time-independent.
- **Time-stepping loop (from $t = 0$ to T):** At each time step, the right-hand side $b(t)$ is updated by evaluating the time-dependent forcing function. The system is then assembled using the previously solved state u^n . Finally, Dirichlet boundary conditions are imposed, the linear system is solved, and the continuous L^2 error over time is accumulated.

Results for theta = 1:

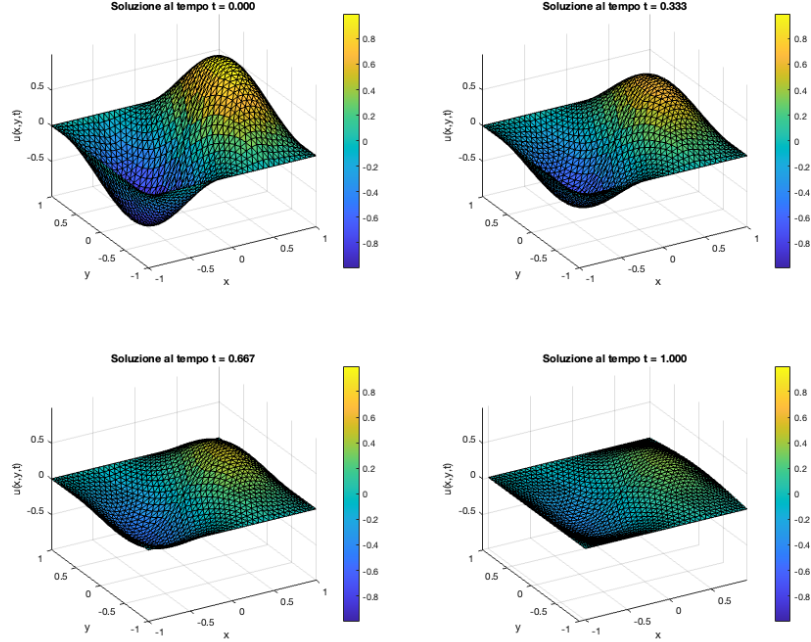


Figure 5: In the figure we notice the dissipative behaviour of the solution as t increases

To compute the space-time global error norm $\|e\|_{L^2(0,T;L^2(\Omega))}$, we approximate the continuous volume integral over each triangular element $T \in \mathcal{T}_h(\Omega)$ using the vertex-based trapezoidal quadrature rule, and accumulate the result over all time steps:

$$\|e\|_{L^2(0,T;L^2(\Omega))}^2 = \int_0^T \|u - u_h\|_{L^2(\Omega)}^2 \approx \sum_{n=1}^{N_{subdiv}} \Delta t \sum_{T \in \mathcal{T}_h(\Omega)} \frac{|T|}{3} \sum_{i=1}^3 \left(u_h^{(n)}(x_i^{(T)}, y_i^{(T)}) - u_{ex}(x_i^{(T)}, y_i^{(T)}, t_n) \right)^2 \quad (24)$$

where $|T|$ represents the area of element T , $(x_i^{(T)}, y_i^{(T)})$ are the local vertices, $u_h^{(n)}$ is the numerical solution at time level n , and u_{ex} is the analytical manufactured solution evaluated at the corresponding time t_n .

3.2.1 Error Estimates and Expected Convergence

The total space-time error $\|e\|_{L^2(0,T;L^2(\Omega))}$ is bounded by a combination of the spatial error from the $\mathcal{P}^1$ Galerkin finite element approximation and the temporal truncation error introduced by the θ -method. From a priori error analysis, we expect the following theoretical bounds:

- **Backward Euler** ($\theta = 1$): This method exhibits first-order accuracy in time and second-order accuracy in space. The theoretical error bound is:

$$\|e\|_{L^2(0,T;L^2(\Omega))} \leq C_1 \Delta t + C_2 h^2. \quad (25)$$

The Backward Euler scheme is unconditionally stable, meaning it does not require any strict time-step constraints (CFL conditions) to prevent numerical blow-up. However, to balance the spatial and temporal errors and achieve an optimal overall convergence rate of $\mathcal{O}(h^2)$, one must couple the time step quadratically to the spatial mesh, i.e., $\Delta t = \mathcal{O}(h^2)$.

- **Forward Euler** ($\theta = 0$): Similar to Backward Euler, this method is first-order in time and second-order in space, sharing the same theoretical error bound:

$$\|e\|_{L^2(0,T;L^2(\Omega))} \leq C_1 \Delta t + C_2 h^2. \quad (26)$$

Unlike the other schemes, Forward Euler is **conditionally stable**. To guarantee convergence and prevent catastrophic exponential growth, the time step must strictly satisfy stability constraint:

$$\Delta t \leq \frac{2}{\lambda_{max}(P^{-1}A)} \leq Ch^2. \quad (27)$$

This rigid stability limit naturally enforces the $\Delta t = \mathcal{O}(h^2)$ coupling required for optimal spatial convergence.

- **Crank-Nicolson** ($\theta = 0.5$): The centered difference evaluation makes this scheme second-order accurate in time. The corresponding theoretical error bound is:

$$\|e\|_{L^2(0,T;L^2(\Omega))} \leq C_1 (\Delta t)^2 + C_2 h^2. \quad (28)$$

Like the Backward Euler method, Crank-Nicolson is unconditionally stable and requires no condition to maintain stability. Furthermore, the error is optimally balanced using only a linear temporal refinement, $\Delta t = \mathcal{O}(h)$. This allows for significantly larger time steps compared to the Euler methods while fully preserving the optimal second-order spatial accuracy.

3.3 Test of Δt : Numerical Results

We will take in consideration only Mesh0, Mesh1, Mesh2 for most of the cases to avoid too small time steps.

3.3.1 $\theta = 1, 1/2$

$\Delta t = 5 * h$:

We use $c=5$ in order to make the first term more dominant.

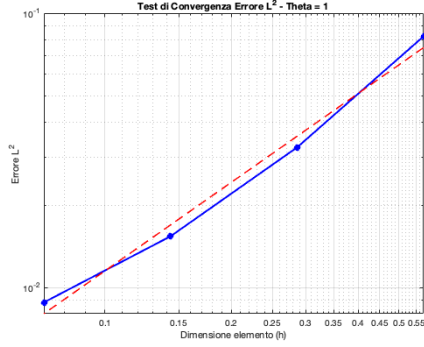


Figure 6: slope ≈ 1.1

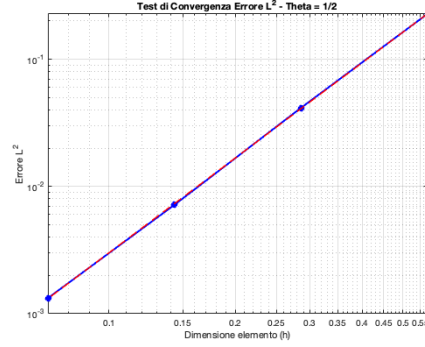


Figure 7: slope ≈ 2.5

As expected, both methods converges even if $\Delta t \approx h$. Moreover the speed of convergence differs as the theory anticipated.

$\Delta t = h^2$:

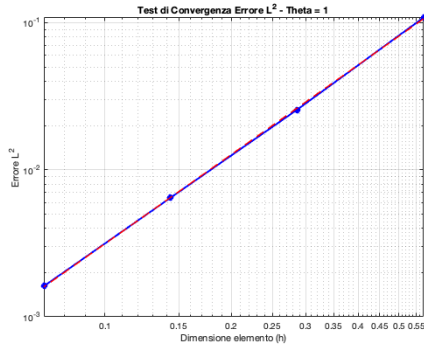


Figure 8: slope ≈ 2

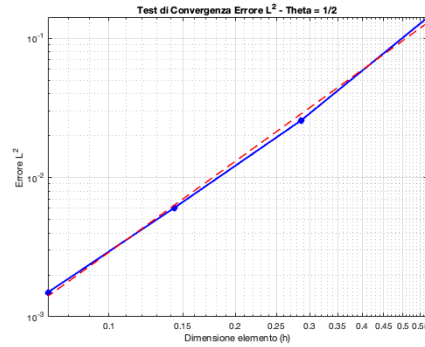


Figure 9: slope ≈ 2.17

Theory estimations satisfied

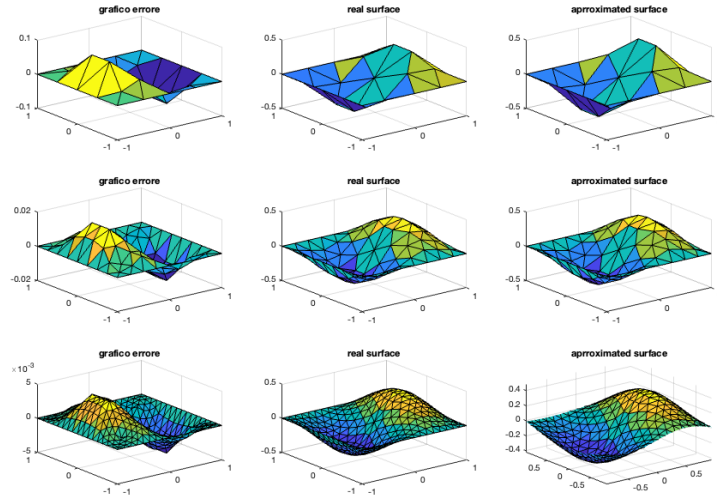


Figure 10: Solution and error surface for $\theta = 1$ and $dt = h^2$

3.3.2 $\theta = 0$

For forward Euler we search for c such that with $\Delta t = c * h^2$ the solution is stable. We will try with $c = [1, 0.1, 0.01]$.

$c = 1$:

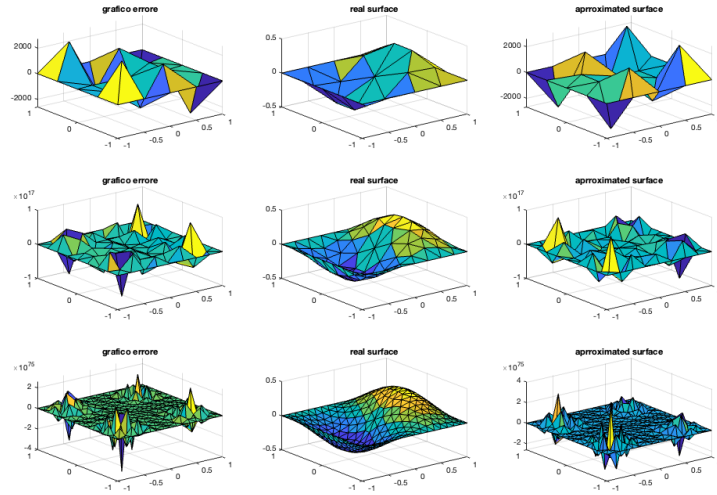


Figure 11: unstable solution

We can understand that the method would not converge for $\Delta t \approx h$ too.

$c = 0.1$:

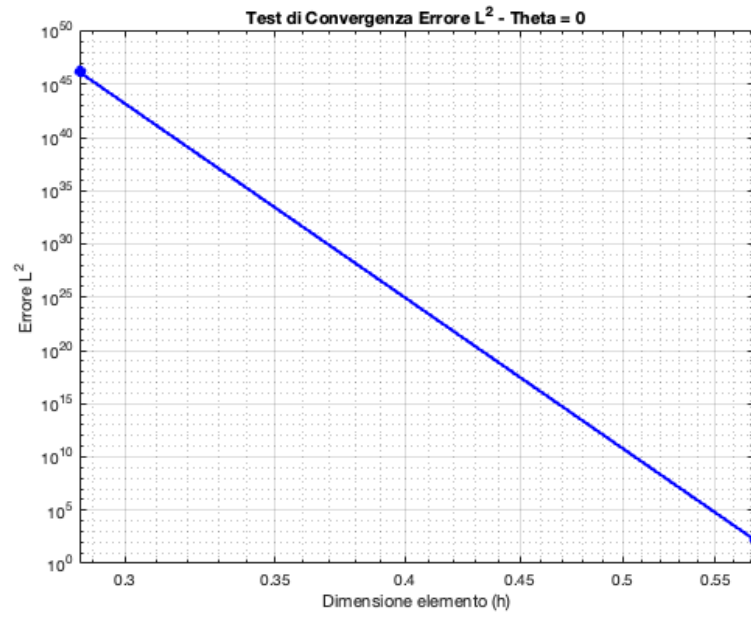


Figure 12: The error is extremely high - unstable solution

$c = 0.01$:

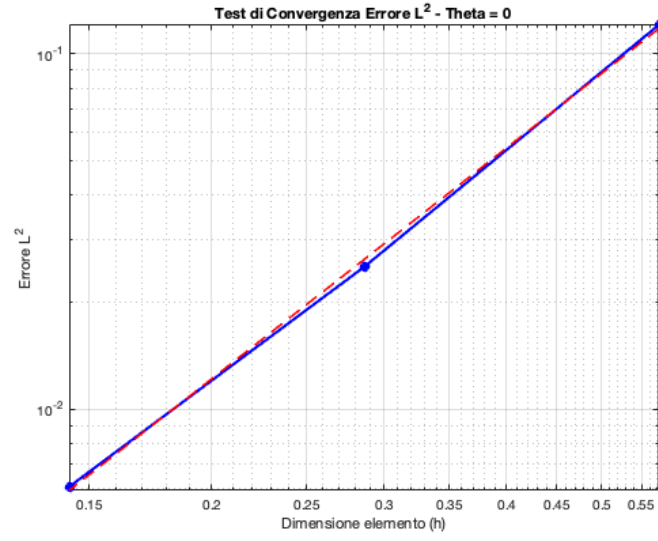


Figure 13: stable convergence

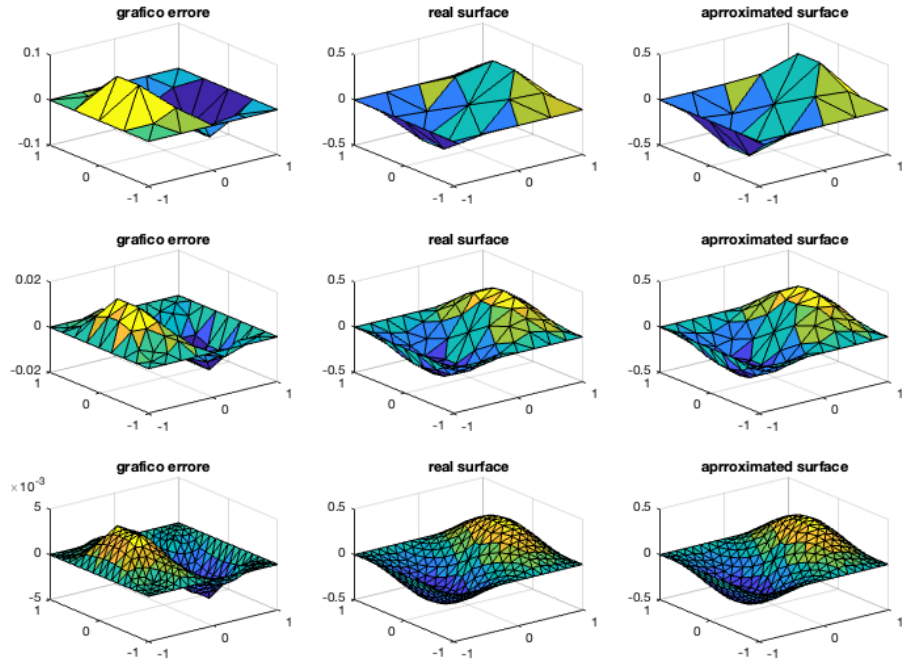


Figure 14: stable solution

Table 2: Number of time steps ($\theta = 0$)

C	Mesh ID	h	N_{steps}	stability
0.01	Mesh 0	0.5725	306	stable
	Mesh 1	0.2863	1221	stable
	Mesh 2	0.131	4882	stable
0.10	Mesh 0	0.5725	31	non stable
	Mesh 1	0.2863	123	non stable
	Mesh 2	0.131	489	non stable
1.00	Mesh 0	0.5725	4	non stable
	Mesh 1	0.2863	13	non stable
	Mesh 2	0.131	49	non stable